ONLINE FOOD DELIVERY DATA ANALYSIS

A look at order trends, revenue, delivery performance and customer behaviour

PROJECT REPORT

I built this dashboard using Python, Pandas, data visualisation and Streamlit. In this report, I walk through the charts in my dashboard, explain the patterns I noticed, and point out where the data needs a little more care before drawing conclusions.

PROJECT SNAPSHOT

98,986 Orders analysed	**■167.55M** Total recorded revenue	**■1,787.10** Average order value
15.0% Cancellation rate	**125 min** Average delivery time	**2.83 / 5** Average delivery rating

Dataset: January–December 2024 | **Coverage:** five cities and five cuisines

Prepared by Lokesh

A quick introduction

For this project, I explored an online food delivery dataset and built an interactive Streamlit dashboard to understand how the business was performing. I used the charts to look at orders, revenue, customer groups, delivery times, ratings, restaurants, payment methods and cancellations.

I have kept the report in the same order as the dashboard, so it is easy to follow along while using the app. For each chart, I explain what I noticed and what I think it tells us. I have also included a few data-quality notes because some patterns can be affected by missing values and the way those values were filled during cleaning.

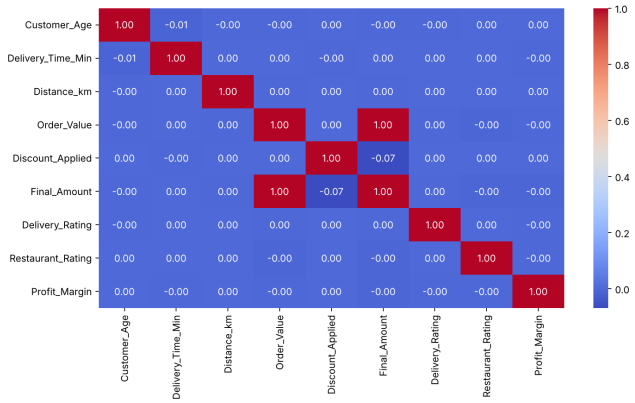
My overall observation: the business looks fairly consistent across the year, but there are a few areas worth investigating. Revenue stays around ■14 million per month, cancellations are close to 15%, delivery takes about two hours on average, and the average delivery rating is 2.83 out of 5.

One thing I would be careful about is treating every chart as a real customer preference. For example, the apparent lead for Card payments, non-peak orders and Indian cuisine is influenced by the missing-value handling. I have marked these points as **Data note** in the relevant sections.

A note on the numbers

Some charts use an axis that does not start at zero. That can make small differences look larger than they really are. I have mentioned this wherever it changes how the chart should be read.

Correlation heatmap



What I noticed

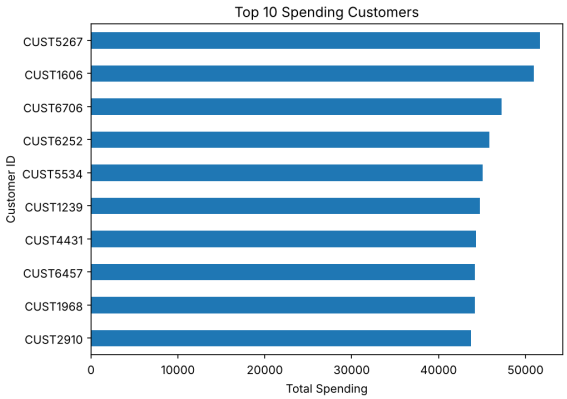
When I looked at the heatmap, most of the values were very close to zero. The clearest relationship was between Order Value and Final Amount (0.997), which makes sense because the final amount is calculated after subtracting the discount. Discount and Final Amount had a very weak negative relationship (-0.07).

What I take from it

Apart from that, the chart does not show meaningful relationships between age, distance, delivery time, ratings and profit margin in this dataset.

> **Data note:** Real delivery data would usually show at least some connection between distance and delivery time. The mostly zero values may be a sign that the data is simulated. Also, many missing ages were filled with the median value of 39, which can reduce variation.

1. Top 10 spending customers



What I noticed

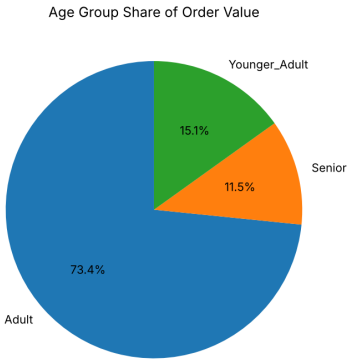
CUST5267 is the highest spender at ■51,677 for the year. The tenth customer on the list, CUST2910, spent ■43,755. The bars are fairly close, so there is no single customer who is far ahead of the rest.

What I take from it

These customers spend around two-and-a-half to three times the typical yearly customer spend of about ■18,000. They are worth retaining, although the top ten together contribute only around 0.3% of total revenue.

> **Data note:** There are roughly 9,000 customers overall, so revenue is spread across a large customer base. Losing one of these top customers would not create a major revenue concentration risk by itself.

2. Age group share of order value



What I noticed

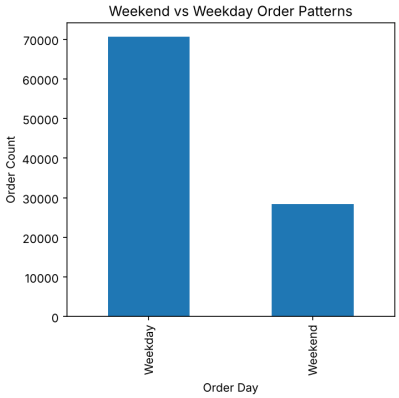
Adults aged 31–50 account for 73.4% of the chart, followed by customers aged 30 and below at 15.1%, and those above 50 at 11.5%.

What I take from it

The adult group also makes up about 73% of orders. The average order value is close to ■1,790 across all age groups, so this chart mainly reflects the number of orders in each group rather than higher spending per order.

> **Data note:** About half of the age values were missing and filled with the median age of 39. Since 39 falls in the adult category, this increases the adult share.

3. Weekend vs weekday order patterns



What I noticed

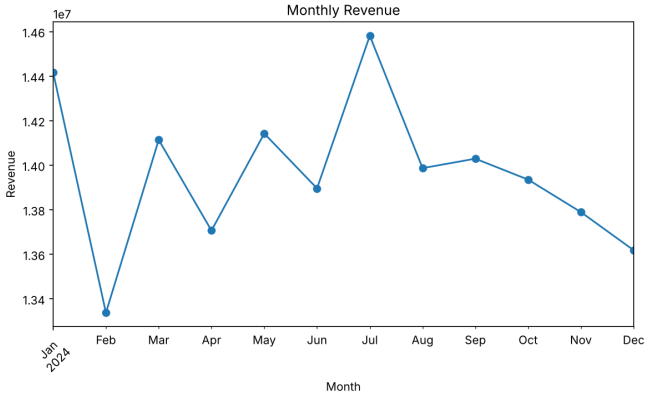
The dashboard shows 70,618 weekday orders and 28,368 weekend orders. At first glance, weekdays seem much busier.

What I take from it

Weekdays cover five days and weekends cover two. When I compare the average per day, it comes to about 14,100 orders on a weekday and 14,200 on a weekend. So daily demand is almost the same.

> **Data note:** A better way to present this chart would be to compare average orders per day instead of total orders.

4. Monthly revenue



What I noticed

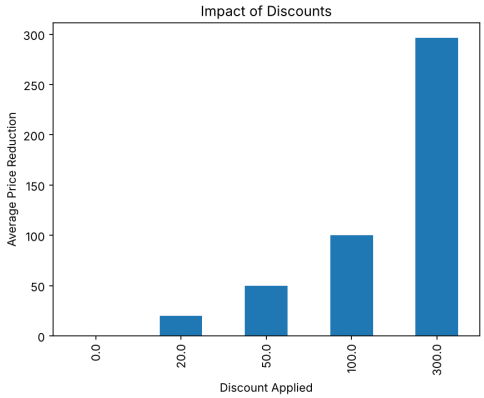
Monthly revenue ranges from about ■13.3 million to ■14.6 million. July is the highest month at ■14.6 million, while February is the lowest at ■13.3 million.

What I take from it

Revenue is quite stable through the year. The difference between the highest and lowest months is around 9%, and there is no strong sign of growth or seasonality. The chart's vertical axis starts above zero, which makes the changes look more dramatic than they are.

> **Data note:** February also has fewer days than most other months, which partly explains its lower total.

5. Impact of discounts



What I noticed

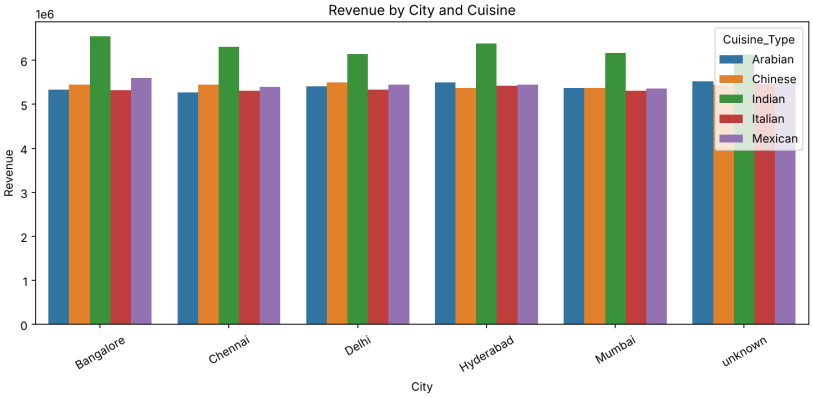
The average price reduction follows the discount amount: ■0, ■20, ■50, ■100 and just under ■300 for the discount groups shown.

What I take from it

This confirms that the discount is being deducted from the order amount, but it does not tell me whether discounts actually increase orders or improve profit. Some smaller orders cannot receive the full ■300 discount because the final amount cannot go below zero.

> **Data note:** To understand whether discounts are useful, I would compare order count, order value and profit margin across the different discount levels.

6. Revenue by city and cuisine



What I noticed

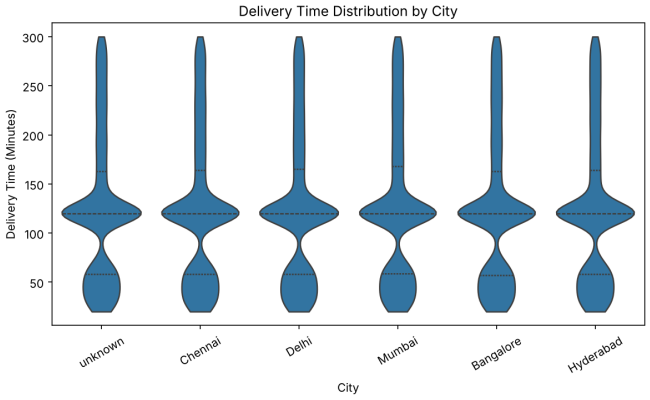
Indian cuisine has the tallest revenue bar in each of the city groups. Bangalore Indian is highest at about ■6.5 million, followed by Hyderabad Indian at ■6.4 million and Chennai Indian at ■6.3 million. Chennai Arabian is the lowest at about ■5.3 million.

What I take from it

Bangalore is slightly ahead overall, but the cities are very close, with total revenue between ■27.6 million and ■28.2 million. The unknown city group also contributes a similar amount.

> **Data note:** The Indian cuisine lead is affected by the way missing cuisine values were filled. I would avoid calling it the clear customer favourite until that is corrected.

7. Delivery time distribution by city



What I noticed

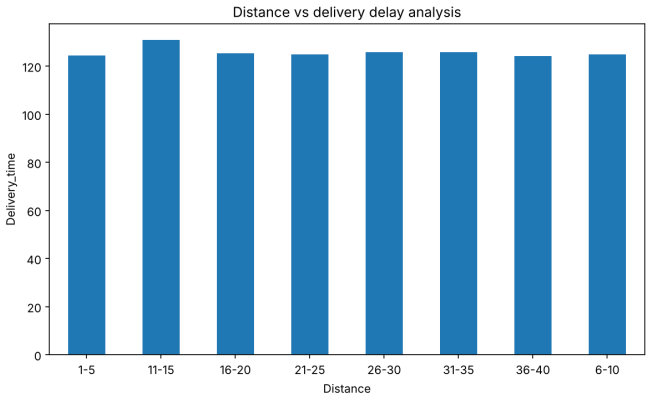
The delivery-time distributions look almost identical across all six city groups. Each one has a large peak around 120 minutes, another smaller one near 55 minutes, and a tail reaching 300 minutes. The median is 120 minutes in every city.

What I take from it

Average delivery time ranges from 124.3 minutes in Bangalore to 126.0 minutes in Mumbai. The differences are very small. The bigger concern is that an average delivery takes just over two hours.

> **Data note:** A large number of missing delivery-time values were filled with the median. This creates the strong peak around 120 minutes and may hide the actual variation.

8. Distance vs delivery delay



What I noticed

The average delivery time is close to 125 minutes across almost every distance group. The 11–15 km group is the highest at about 131 minutes.

What I take from it

The chart does not show a clear increase in delivery time as distance increases. In a real delivery operation, I would expect distance to have at least some effect. Preparation time, rider availability or limitations in the dataset may explain this pattern.

Data note: The distance groups are currently sorted like text, so the order on the chart is not correct. I would set them as an ordered category so the groups appear from shortest to longest distance.

9. Delivery rating vs delivery time



What I noticed

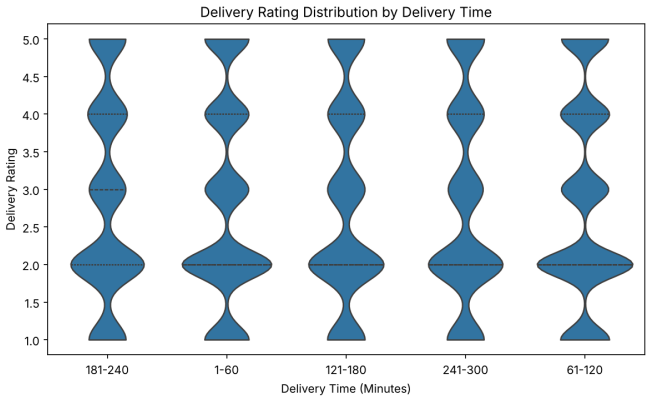
The rating distributions look very similar across all delivery-time groups. Ratings from 1 to 5 appear in each group, and the widest part of the distributions is around rating 2. Average ratings are between 2.82 and 2.84.

What I take from it

The chart suggests that ratings do not change much with delivery time in this dataset. That is unusual because delivery speed often affects customer satisfaction. It also means I cannot use this chart alone to explain the low overall rating.

Data note: Missing ratings were filled with the most common value, 2, which makes the peak at 2 larger. The delivery-time groups also need to be sorted in the correct order.

10. Top 10 restaurants by average rating



What I noticed

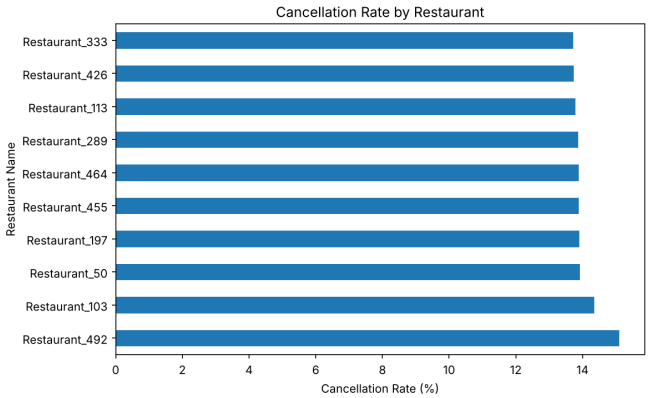
Restaurant_101 has the highest average rating at 4.33, followed very closely by Restaurant_162 at 4.32 and Restaurant_1 at 4.31. The tenth restaurant has an average of 4.30.

What I take from it

The difference between the top ten is only about 0.03 points. They are practically tied, so I would not say that Restaurant_101 is clearly better than the others based on this chart alone.

Data note: The chart's axis starts around 4.295 rather than zero, making the small differences look bigger. Ratings above 5 in the original data, up to 5.5, were capped at 5 during cleaning.

11. Cancellation rate by restaurant



What I noticed

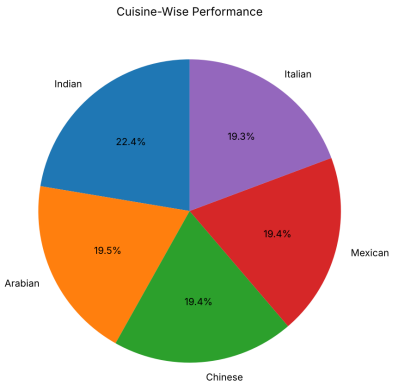
Restaurant_492 has the highest displayed cancellation rate at 15.1%. Restaurant_103 follows at 14.3%, while the remaining restaurants are close to 13.7–13.9%.

What I take from it

These are the ten highest out of around 500 restaurants, and each has roughly 200 orders. A few restaurants will naturally appear at the top when so many are compared. I would keep them on a watch list rather than blame one restaurant immediately.

Data note: This chart counts cancellations only when a cancellation reason is recorded. Around 5,942 cancelled orders have no reason, so the chart understates the actual cancellation rate. The Order Status column is a better source for this calculation.

12. Cuisine-wise performance



What I noticed

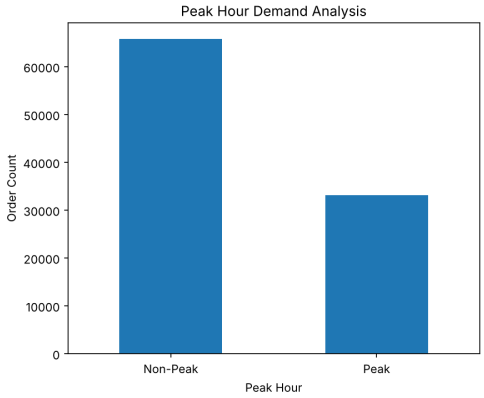
Indian cuisine accounts for 22.4% of orders. Arabian, Chinese, Mexican and Italian each account for roughly 19.4%.

What I take from it

Indian appears to be the leading cuisine in this chart, and its revenue is about ■37.7 million compared with around ■32–33 million for the others. However, the order share is affected by missing-value handling.

Data note: In the original data, the five cuisines were almost evenly represented at around 20% each. Filling missing cuisine values with the most common cuisine in each area added roughly 5,500 orders to Indian. So I would not present this as a genuine customer preference.

13. Peak hour demand analysis



What I noticed

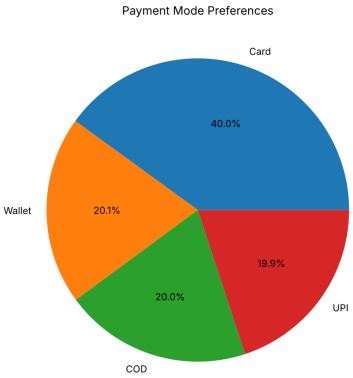
The dashboard shows 65,845 non-peak orders and 33,141 peak-hour orders. That makes non-peak demand look almost twice as high.

What I take from it

This result is surprising for a food delivery business. When I checked the original data, peak and non-peak orders were almost equal: 33,585 non-peak and 33,453 peak orders.

Data note: Around one-third of the peak-hour values were missing and were filled as False, which classified them as non-peak. This created the difference shown in the dashboard.

14. Payment mode preferences



What I noticed

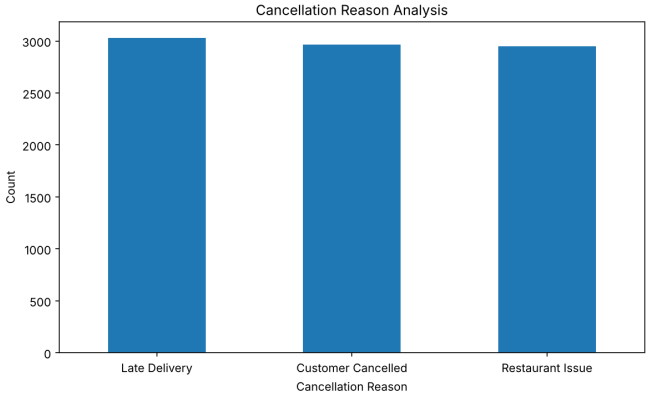
Card payments make up 40% of the orders, while Wallet, COD and UPI each account for about 20%.

What I take from it

At first, this makes Card look like the clear favourite. But the original data has all four payment modes at almost the same level, close to 20,000 orders each.

Data note: Around 19,700 missing payment modes were filled with Card. Because of that, I cannot confidently say that customers prefer Card payments.

15. Cancellation reason analysis



What I noticed

The three recorded reasons are very close in count: Late Delivery has 3,032 cases, Customer Cancelled has 2,963, and Restaurant Issue has 2,949.

What I take from it

There is no single reason that clearly dominates. Late Delivery is slightly higher, but the difference is small. This suggests that cancellations may need to be addressed through more than one improvement.

Data note: Only 8,944 of the 14,886 cancelled orders have a reason recorded. The remaining 5,942 are marked as Not Applicable, so this chart explains about 60% of cancellations.

My overall takeaways

After going through all the charts, these are the points I would be comfortable presenting:

• Revenue stays fairly steady at around ■14 million per month, with July showing the highest value.
• Around 15% of orders are cancelled.
• Average delivery time is about 125 minutes across cities.
• The average delivery rating is 2.83 out of 5, which leaves room for improvement.
• Weekday and weekend order volume is almost the same when compared per day.
• Revenue is quite evenly spread across cities and cuisines.

What I would be careful about

Some dashboard insights are influenced by the way missing values were handled. In particular, the apparent preference for Card payments, the higher non-peak order count and the lead for Indian cuisine should not be treated as confirmed customer behaviour. The adult age-group share is also affected by filling missing ages with the median. Bangalore is slightly ahead in revenue, but the difference between cities is small.

What I would improve next

1. Keep missing payment mode, peak-hour and cuisine values as “Unknown” instead of filling them with one category, then update those charts.
2. Add a flag to identify values that were filled during data cleaning.
3. Calculate the cancellation rate using Order Status, and show revenue from delivered orders separately. Cancelled orders currently contribute about ■25.1 million to the revenue total.
4. Compare weekday and weekend orders per day, and sort the distance and delivery-time groups in the correct order.
5. Compare order volume and profit margin across discount levels to understand whether discounts are actually helping the business.

Final note

Building this dashboard helped me practise data cleaning, exploratory data analysis, visualisation and presenting insights through Streamlit. It also reminded me that a chart is only as reliable as the data behind it. With the data-quality improvements above, I can make the dashboard's conclusions more useful and easier to defend in a presentation.